Decoding the Little Man Computer

Bridging the Handwritten Notes to the Emulator

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The Architecture

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Investigate the Emulator:

Look at the Peter Higginson website.

- How many input and output channels do you see?
- Do you see any external storage like a hard drive or flash memory represented?

Clue 1: Mnemonics vs. Machine Code

Look at your handwritten table under the columns **MNEMONIC** and **MLBOX/OP**.

You have specific instructions like:

- `Load LDA (5)`
- `Store STO (3)`
- `Halt HLT (0)`

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Your Task:

In the emulator's RAM, each mailbox holds exactly 3 digits. If the Operation Code (OP) for `LDA` is 5, how does a computer combine the instruction "5" and a mailbox address "12" to make a single piece of machine code?

Clue 2: The "RESULT" Column

In your handwritten notes, look at mailbox `001`.

The table shows the mnemonic add `ADD (1)` acting on `1 15 (390)`, and the RESULT column shows = `391`.

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Investigate the Math:

- If the operation at 001 adds 390 to get 391, what value was already waiting inside the CPU?
- Run an addition program in the emulator and click **STEP** instead of RUN. Which specific box in the CPU updates to show the "RESULT" from your paper?

Look at the very bottom of your handwritten notes at mailbox `010`.

The mnemonic is `BRANCH BRP(8) POSITIVE`, and there is a physical arrow drawn from that line pointing back up to the top.

Clue 3: The Flow of Control

Look at the very bottom of your handwritten notes at mailbox `010`.

The mnemonic is `BRANCH BRP(8) POSITIVE`, and there is a physical arrow drawn from that line pointing back up to the top.

Connect to Past Labs:

- Step through a program in the emulator until a `BRP` instruction fires. Watch the **Program Counter**. What happens to it?
- Your lab manual asks you to look at the "bubble sort" program. Does this branching arrow have ANY connection to the practice Linux shell scripts you saw in Lab 5?

Your Self-Study Mission

Your goal today is not to just run the code, but to understand the machine's logic to prepare for your Moodle Quiz.

To prepare:

1. Load the **add** or **bubble sort** code into the emulator.
2. Slow the execution speed down using the **OPTIONS** menu.
3. Track one piece of data. Watch it move from the **INPUT**, to the CPU, into a **RAM Mailbox**, and eventually to the **OUTPUT**.

If you can trace the data on the screen, you can map it to your handwritten notes.